```
#EL OTMANI Youssef
#1
R = PolynomialRing(QQ, ['x', 'y'], order='lex')
#x, y = R.gens()
x=var('x')
y=var('y')

#2
def spol(f, g):
    t = lcm(f.lm(), g.lm())
    return ((g.lc() * t * f) // f.lm()) - ((f.lc() * t * g) // g.lm())

#3
f1 = R(x^2 - 5*y^3 - 5)
f2 = R(x^2 + y^4 - 5*y^2)

F=[f1,f2]

#4
p1 = implicit_plot(f1, (x, -5, 5), (y, -5, 5), color='blue')
p2 = implicit_plot(f2, (x, -5, 5), (y, -5, 5), color='red')
show(p1 + p2)

#5 intersection : 2 points,  solution visually  = (-1.75, -0.75) (1.75, -0.75)


#6
h = spol(f1, f2)
print("S-polynomial:", h)

#7

#8
h_reduced = h.reduce(F)
print("h reduced:", h_reduced)
#not reducucible

#6
F.append(h)
print(F)

h1 = spol(f1, h)
h2 = spol(f2, h)

h1_reduced = h1.reduce(F)
h2_reduced = h2.reduce(F)

print("h1 reduced:", h1_reduced)
print("h2 reduced:", h2_reduced)
# we can see that h1 and h2 can be reduced using F
#So F is a grobner basis

#10
print("f2 -f1 + h =", f2 -f1 + h )
old_F = F
F = [f1, -h]
print(F)

#11
print("are the ideals the same ?", Ideal(F)==Ideal(old_F) )

#12
print("Gröbner basis:", ideal([f1, f2]).groebner_basis())
print("our reduced Gröbner basis:", F)
#we can see that they are the same, so {f1,h} is a reduced grobner basis


#7
for poly in F:
    if poly.variables() == (y,):
        poly_y = poly
        break

factored_poly_y = poly_y.factor()
```

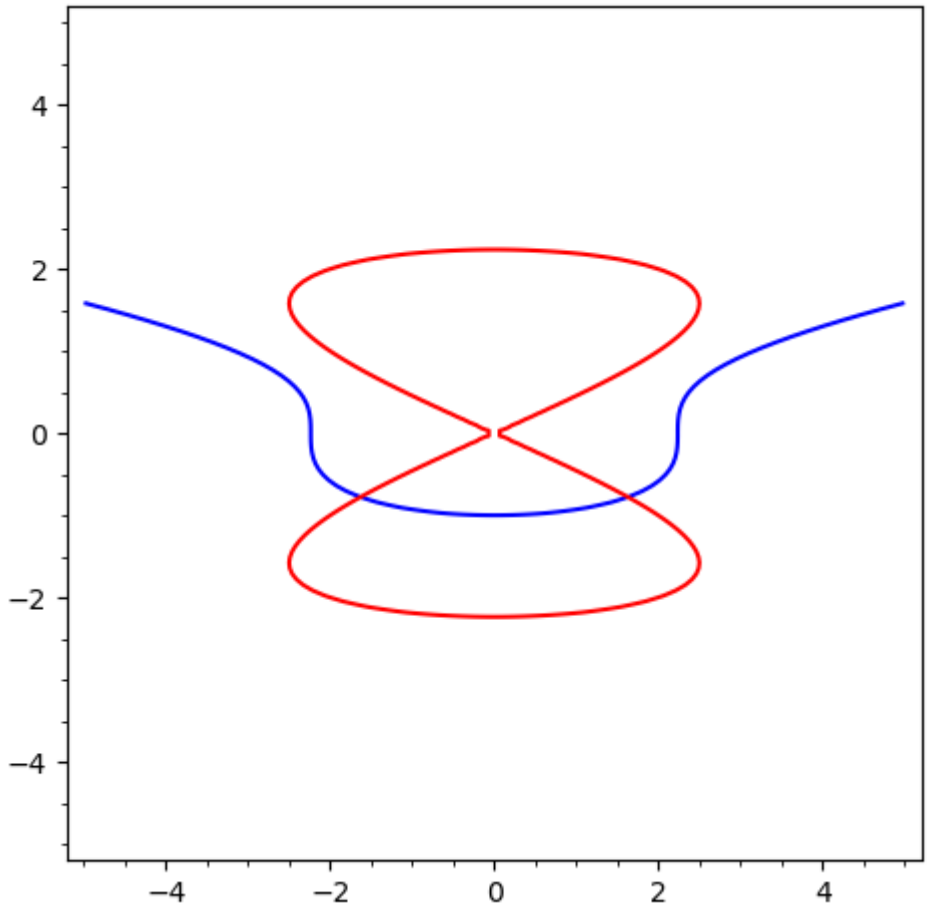
```
print("Factorized polynomial in y:", factored_poly_y)


eq = y^4 + 5*y^3 - 5*y^2 + 5 == 0
y_solutions = solve(eq, y)
print("there is ", len(y_solutions), "solutions for y")
print("Solutions for y:", y_solutions)
#we see that the solutions are real solutions


x_solutions = []
for sol in y_solutions:
    y_val = sol.rhs()
    x_eq = f2.subs(y=y_val)
    x_solutions.append(solve(x_eq, x))

count = 0
for i, sol in enumerate(y_solutions):
    print("is the y solution real ?", sol.rhs().is_real())
    print(f"For y = {sol.rhs()}, x solutions, wich there are {len(x_solutions[i]) },  are: {x_solutions[i]}\n")
    count += len(x_solutions[i])

print("so " , count, "(x,y) solutions")
```


4
2
0
−2
−4
−4
−2
0
2
4

```
Oout :
S-polynomial: -y^4 - 5*y^3 + 5*y^2 - 5
h reduced: -y^4 - 5*y^3 + 5*y^2 - 5
[x^2 - 5*y^3 - 5, x^2 + y^4 - 5*y^2, -y^4 - 5*y^3 + 5*y^2 - 5]
h1 reduced: 0
h2 reduced: 0
f2 -f1 + h = 0
[x^2 - 5*y^3 - 5, y^4 + 5*y^3 - 5*y^2 + 5]
are the ideals the same ? True
Gröbner basis: [x^2 - 5*y^3 - 5, y^4 + 5*y^3 - 5*y^2 + 5]
our reduced Gröbner basis: [x^2 - 5*y^3 - 5, y^4 + 5*y^3 - 5*y^2 + 5]
Factorized polynomial in y: y^4 + 5*y^3 - 5*y^2 + 5
there is  4 solutions for y
Solutions for y: [
y == -1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^(1/
y == -1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^(1/
y == 1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^(1/3
y == 1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^(1/3
]
is the y solution real ? False
For y = -1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^
x == -1/9*sqrt(-153/20*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)^(3/2)*
x == 1/9*sqrt(-153/20*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)^(3/2)*s
]

is the y solution real ? False
For y = -1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^
x == -1/24*sqrt(1/5)*sqrt(272*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)
x == 1/24*sqrt(1/5)*sqrt(272*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)^
]

is the y solution real ? False
For y = 1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^(
x == -1/24*sqrt(1/5)*sqrt(272*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)
x == 1/24*sqrt(1/5)*sqrt(272*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)^
]

is the y solution real ? False
For y = 1/12*sqrt(6/5)*sqrt((5/2)^(2/3)*(4*(5/2)^(2/3)*(9*sqrt(10765) + 985)^(2/3) + 115*(5/2)^(1/3)*(9*sqrt(10765) + 985)^(1/3) + 340)/(9*sqrt(10765) + 985)^(
x == -1/9*sqrt(-153/20*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)^(3/2)*
x == 1/9*sqrt(-153/20*sqrt(6)*sqrt(5)*2^(2/3)*(5*5^(1/3)*2^(2/3)*(9*sqrt(10765) + 985)^(1/3) + 170*5^(2/3)*2^(1/3)/(9*sqrt(10765) + 985)^(1/3) + 575/2)^(3/2)*s
]

so  8 (x,y) solutions
```

```
IN :
R = PolynomialRing(QQ, ['w','x', 'y', 'z'], order='lex')
x=var('x')
y=var('y')
w=var('w')
z=var('z')

def spol(f, g):
    t = lcm(f.lm(), g.lm())
    return ((g.lc() * t * f) // f.lm()) - ((f.lc() * t * g) // g.lm())

f1 = R(w^2 -x^2 - 3*y^2 + z^4 - z^2 -3)
f2 = R(w^2 - 4*y^2 - 6*z^2 - 2)
f3 = R(2*w^2 -2*x^2 - 6*y^2 - 2*z^2 -4)
f4 = R(w^2 -2*x^2 - 3*y^2 + z^4 + 5*z^2)


F=[f1,f2, f3, f4]

print(len(F))

for i in range(len(F)):
    for j in range(i + 1, len(F)):
        s_poly = spol(F[i], F[j])
        s_reduced = s_poly.reduce(F)
        if(s_reduced != 0):
            F.append(s_poly)
            i=1
            break

print(len(F))

#there are a lot of reduction needed so we will just use the sage grobner basis function
G = ideal([f1, f2, f3, f4]).groebner_basis()


print(G)
for poly2 in G:
    if poly2.variables() == (z,):
        poly_z = poly2
        break

factored_poly_z = poly_z.factor()
print("Factorized polynomial in z:", factored_poly_z)

eq = (z - 1) * (z + 1) * (z^2 + 1) == 0
z_solutions = solve(eq, z)
print("there is ", len(z_solutions), "solution for z")
print("Solutions for z:", z_solutions)

for sol in z_solutions:
    if(sol.rhs().is_real() ):
       z_val = sol.rhs()
       f1_eq = f1.subs(z=z_val)
       f2_eq = f2.subs(z=z_val)
       f3_eq = f3.subs(z=z_val)
       f4_eq = f4.subs(z=z_val)
       print("Équations après substitution :(1)", f1_eq, "(2)", f2_eq,"(3)", f3_eq,"(4)", f4_eq)

#same equations for both Z solutions

R = PolynomialRing(QQ, ['w', 'x', 'y'], order='lex')
x=var('x')
y=var('y')
w=var('w')
```

```
f1_poly = R(f1_eq)
f2_poly = R(f2_eq)
f3_poly = R(f3_eq)
f4_poly = R(f4_eq)

G = ideal([f1_poly, f2_poly, f3_poly, f4_poly]).groebner_basis()

print("Gröbner after substitution :", G)
nb =0
eq1 = w^2 - 24 ==0
eq2 = x^2 - 9 ==0
eq3 = y^2 - 4 ==0
for sol in z_solutions :
    if(sol.rhs().is_real() ):
        for sol2 in solve(eq1,w) :
            for sol3 in solve(eq2,x) :
                for sol4 in solve(eq3,y) :
                    print( sol,  sol2,  sol3, sol4)
                    nb +=1

print("nb of solutions ", nb)
```

```
Out:
4
8
[w^2 - 10*z^2 - 14, x^2 - 6*z^2 - 3, y^2 - z^2 - 3, z^4 - 1]
Factorized polynomial in z: (z - 1) * (z + 1) * (z^2 + 1)
there is  4 solution for z
Solutions for z: [
z == -I,
z == I,
z == -1,
z == 1
]
Équations après substitution :(1) w^2 - x^2 - 3*y^2 - 3 (2) w^2 - 4*y^2 - 8 (3) 2*w^2 - 2*x^2 - 6*y^2 - 6 (4) w^2 - 2*x^2 - 3*y^2 + 6
Équations après substitution :(1) w^2 - x^2 - 3*y^2 - 3 (2) w^2 - 4*y^2 - 8 (3) 2*w^2 - 2*x^2 - 6*y^2 - 6 (4) w^2 - 2*x^2 - 3*y^2 + 6
Gröbner after substitution : [w^2 - 24, x^2 - 9, y^2 - 4]
z == -1 w == -2*sqrt(6) x == -3 y == -2
z == -1 w == -2*sqrt(6) x == -3 y == 2
z == -1 w == -2*sqrt(6) x == 3 y == -2
z == -1 w == -2*sqrt(6) x == 3 y == 2
z == -1 w == 2*sqrt(6) x == -3 y == -2
z == -1 w == 2*sqrt(6) x == -3 y == 2
z == -1 w == 2*sqrt(6) x == 3 y == -2
z == -1 w == 2*sqrt(6) x == 3 y == 2
z == 1 w == -2*sqrt(6) x == -3 y == -2
z == 1 w == -2*sqrt(6) x == -3 y == 2
z == 1 w == -2*sqrt(6) x == 3 y == -2
z == 1 w == -2*sqrt(6) x == 3 y == 2
z == 1 w == 2*sqrt(6) x == -3 y == -2
z == 1 w == 2*sqrt(6) x == -3 y == 2
z == 1 w == 2*sqrt(6) x == 3 y == -2
z == 1 w == 2*sqrt(6) x == 3 y == 2
nb of solutions  16
```